

2.

[2]

- (b) An athlete uses a machine in the gym where he hinges at the knee joint to move his lower legs up as shown in Fig. 1.1. There is a constant downward force of 25 N exerted at point A on each feet at constant distance L from point B, as shown in Fig. 1.2. The combined weight of a foot and one lower leg is 30 N acting at $L/2$ from the knee joint. When the legs are raised, there is a force F at distance $L/4$ from knee joint and at angle of θ to the leg as shown in Fig 1.2.

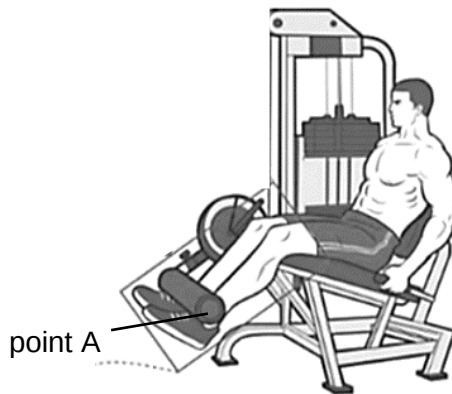


Fig. 1.1

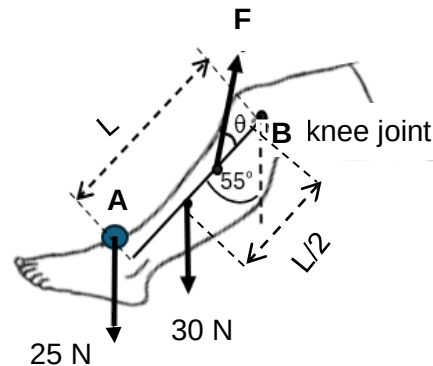


Fig 1.2

It is given that $L = 40$ cm. At position shown in Fig. 1.2, the lower leg and feet are at equilibrium at angle of 55° to the vertical, and $\theta = 25^\circ$.

- (i) By taking moments about point B at the knee joint, calculate the force F exerted on the lower leg in Fig 1.2.

force, $F = \dots\dots\dots$ N [2]

[Turn over]

- (ii) For the lower leg to be in equilibrium, there is a force R at the knee joint.

Draw on Fig.1.2, a labelled arrow to represent the force R .

[1]

- (iii) At the beginning, the feet are down as shown in Fig. 1.3.

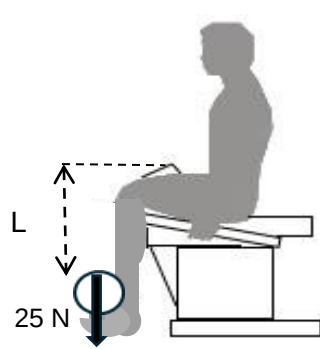


Fig 1.3

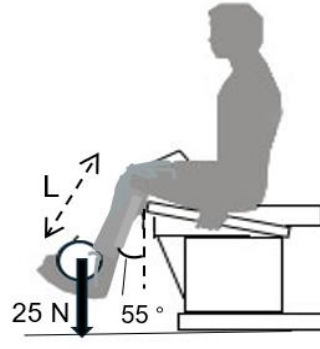


Fig 1.4

He raises his legs until they are at an angle of 55° to the vertical as shown in Fig. 1.4. Calculate the work done to raise one lower leg to this position at constant speed. Assume that the knee joint is a point that stays in place throughout.

[2]

- (iv) Hence calculate the power exerted by the athlete in raising one leg to the position shown in Fig 1.4 if the time taken is 5.0 s.

Power exerted = W [1]

- 2 (a) (i) Explain why the gravitational potential at a point in a gravitational field is negative.

.....

 [2]

- (ii) The gravitational potential at the surface of Earth is $-62.6 \times 10^6 \text{ J kg}^{-1}$, and that at the surface of Moon is $-28.1 \times 10^6 \text{ J kg}^{-1}$. (The mass of the Earth is 81 times the mass of the Moon).

1. On Fig. 2.1, sketch a graph which shows the variation of gravitational field strength along a line from the surface of Earth to the surface of Moon. [2]
2. Hence sketch, on Fig. 2.2, a graph which shows the variation of gravitational potential along a line from the surface of Earth to the surface of Moon. [2]

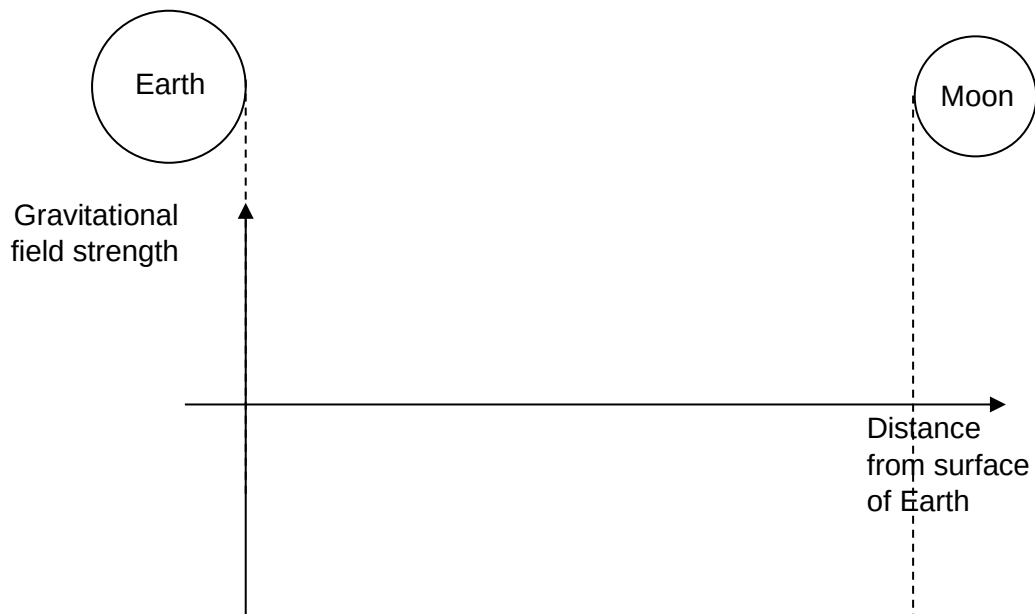


Fig. 2.1



Fig. 2.2

- (b) An isolated spherical planet has a diameter of 6.8×10^6 m. Its mass of 6.4×10^{23} kg may be assumed to be a point mass at the centre of the planet.

(i) Show that the gravitational field strength at the surface of the planet is 3.7 N kg^{-1} . [1]

(ii) A stone of mass 2.4 kg is raised from the surface of the planet through a vertical height of 1800 m. Use the value of the field strength from (i) to determine the change in gravitational potential energy of the stone. Explain your working.

change in gravitational potential energy = J [2]

(iii) A rock, initially at rest at infinity, moves towards the planet. At point P, its height above the surface of the planet is $3.5 D$, where D is the diameter of the planet, as shown in Fig. 2.3.

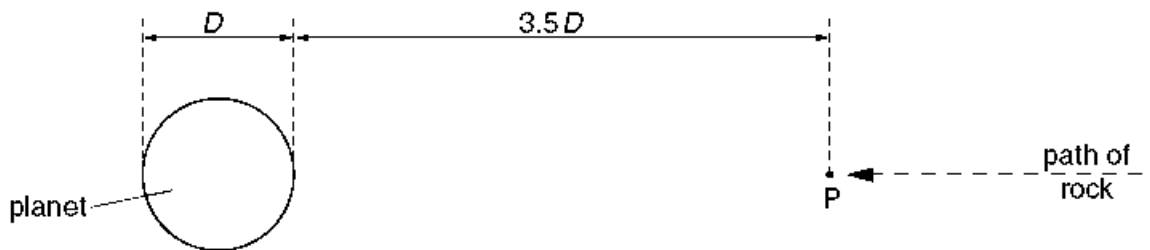


Fig. 2.3

Calculate the speed of the rock at point P. Explain your working,

speed at point P = m s^{-1} [3]

